Lab report

Structural Characterisation

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Experiment	Imaging and electron diffraction of TEM

Experiment II Imaging and electron diffraction of TEM

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Grade	Third				
Contents	1. Working principle of transmission electron microscope 2. Analysis of TEM contrast 3. Calibration of electron diffraction pattern				

1. Working principle of transmission electron microscope. (30 marks)

A transmission electron microscope (TEM) employs a high-energy electron beam to probe ultrathin specimens and achieve atomic-scale structural resolution. Its operation follows a coherent physical process in which elastically scattered electrons are encoded with specimen information and subsequently magnified by electromagnetic lenses.

A. Electron Beam Generation and Conditioning

In an ultrahigh-vacuum environment, a field-emission gun (FEG) emits a highly coherent, high-brightness electron beam, which is then accelerated to high voltage, yielding an extremely short de Broglie wavelength. Condenser lenses demagnify and shape the source to produce a controlled, nearly parallel illumination on the specimen.

B. Interaction with the Specimen

As the beam transmits through the ultrathin sample, electrons undergo elastic and inelastic scattering. The elastically scattered electrons carry essential structural information—their angular distribution and intensity reflect local atomic number, crystal structure, and thickness—thereby encoding microstructural details into the amplitude and phase of the exit wave.

C. Image Formation Based on Abbe's Principle

The exit wave is focused by the objective lens. In its back focal plane, Fraunhofer diffraction produces the diffraction pattern (reciprocal-space information). In the image plane, interference of diffracted beams forms the first real, magnified image. Inserting an objective aperture allows contrast control: selecting the transmitted beam generates bright-field (BF) images, while selecting a diffracted beam yields dark-field (DF) images.

D. Mode Switching and Signal Detection

Intermediate and projector lenses further magnify the image or diffraction pattern. Adjusting the intermediate lens enables rapid switching between imaging and diffraction modes. A selected-area (SA) aperture isolates a defined region of the specimen, allowing direct correlation between morphology and crystallographic structure. Finally, the electron signal is recorded by a fluorescent screen or CCD detector.

2. Analysis of TEM contrast. (30 marks)

Contrast is fundamental to transmission electron microscopy (TEM), referring to the light–dark variations that arise from changes in the intensity or phase of electrons transmitted through the specimen. These variations originate from differences in how regions of the sample scatter or phase-shift the incident electron beam. In TEM, three primary contrast mechanisms are involved:

(1) Thickness contrast: Dominant in amorphous or nanostructured materials, produced by variations in specimen thickness and atomic number.

(2) Diffraction contrast: Characteristic of crystalline materials, governed by electron diffraction, and sensitive to crystal orientation, strain, and defects.

(3) Phase contrast: Used in high-resolution TEM, formed by phase interference between the transmitted and diffracted beams, enabling atomic-scale structural imaging.

A. The First TEM Image.

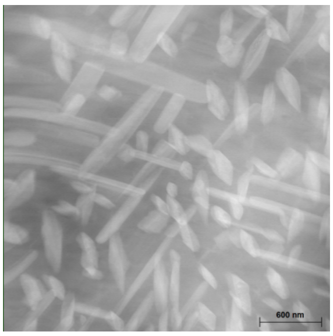

Figure 1. The first TEM image.

Result: The TEM image is a **dark-field one with diffraction contrast**. Numerous needle-like crystals show bright contrast against a dark background, featuring regular morphology, uniform distribution, and nanoscale size (hundreds of nanometers, scale bar indicated).

Reason: This contrast arises from electron diffraction and the selective filtering of the objective aperture. During imaging, the objective aperture blocks the transmitted beam and allows only a specific diffracted beam that satisfies the Bragg condition to form the image. The needle-like crystals meet this condition and thus produce bright contrast, whereas regions that do not generate the corresponding diffracted beam appear dark. This mechanism reveals differences in crystal orientation and crystallinity, reflecting the presence of phase-transformation products or nanoscale precipitates.

B. The Second TEM Image.

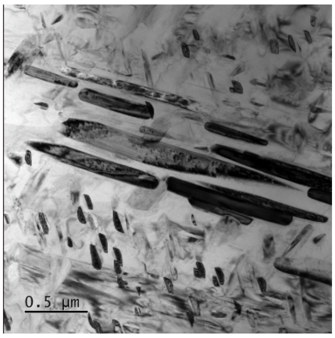

Figure 2. The second TEM image.

Result: This image is a **bright-field one with diffraction contrast**. Numerous elongated and granular crystals appear dark in diffraction-satisfying regions and bright in the background/non-diffracting regions, featuring diverse morphologies, regular distribution, and submicron size (hundreds of nanometers, scale bar indicated).

Reason: The bright-field image arises from electron diffraction combined with the filtering effect of the objective aperture. In bright-field mode, only the transmitted beam is allowed to form the image. When elongated or granular crystals satisfy specific Bragg diffraction conditions, a substantial portion of electrons is diffracted and blocked by the aperture, reducing the transmitted intensity and producing dark contrast. Regions that do not meet the diffraction condition transmit electrons efficiently and therefore appear bright. This contrast mechanism reflects differences in crystal orientation and crystallinity, indicating the presence of multiscale secondary phases—such as transformation products or precipitates—and provides a basis for analyzing their morphology, distribution, and crystallographic features.

C. The Third TEM Image.

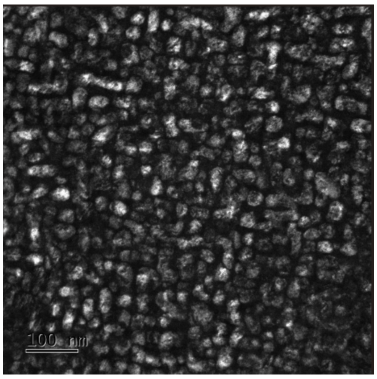

Figure 3. The third TEM image

Result: The contrast type of this image is **mass-thickness contrast**. The image contains numerous nanoscale particles, with the particles showing bright contrast and the background appearing dark. The particles are approximately spherical in morphology, relatively uniform in distribution, and at the nanoscale (with a characteristic size of tens of nanometers as indicated by the scale bar).

Reason: The formation of mass–thickness contrast arises from differences in atomic number and specimen thickness, which lead to variations in electron-scattering ability between the nanoscale particles and the surrounding matrix. Regions containing particles with higher atomic number or greater thickness scatter electrons more strongly, causing a larger fraction of high-angle electrons to be blocked by the objective aperture. As a result, fewer transmitted electrons contribute to imaging in these areas, producing bright contrast, while the background appears darker due to weaker scattering. This contrast behavior indicates the presence of uniformly distributed nanoscale particle phases—such as nanoalloy particles or quantum dots—and serves as a characteristic imaging signature for analyzing their size, distribution, and compositional variations.

3. Calibration of electron diffraction pattern. (40 marks)

The radius R of each diffraction spot is the key parameter for indexing. In selected-area electron diffraction, the interplanar spacing d is related to R through:

$$d = \frac{L\lambda}{R}$$

For a cubic crystal system, the relationship between the interplanar spacing d, the lattice constant a, and the Miller indices (H K L) is given by:

$$d = \frac{a}{\sqrt{H^2 + K^2 + L^2}}$$

In the cubic crystal system, the magnitude of the reciprocal lattice vector g corresponding to the crystal plane (H K L) is:

$$\frac{1}{|g|} = \frac{\sqrt{H^2 + K^2 + L^2}}{a} = \frac{1}{d}$$

Where a is the lattice constant and d is the interplanar spacing. Therefore, the sum of the squares of the Miller indices $N = H^2 + K^2 + L^2$ is proportional to $1/d^2$.

We know that R1=6 mm, R2=9.8 mm, R3=11.5 mm θ=31.4°, Therefore, it can be calculated that:

$$N_1 : N_2 : N_3 = R_1^2 : R_2^2 : R_3^2 = 6^2 : 9.8^2 : 11.5^2 \approx 3 : 8 : 11$$

Suppose that:

$$\overrightarrow{g_1} = (1\ \bar{1}\ 1), \qquad \overrightarrow{g_2} = (0\ 2\ 2)$$

Then，

$$\overrightarrow{g_3} = (3\ 1\ 1),\ |\overrightarrow{g_1}| = \sqrt{3},\ |\overrightarrow{g_2}| = \sqrt{8},\ |\overrightarrow{g_3}| = \sqrt{11}$$

$$\cos\theta = \frac{\overrightarrow{g_2} \cdot \overrightarrow{g_3}}{|\overrightarrow{g_2}||\overrightarrow{g_3}|} = \frac{0 \times 3 + 2 \times 1 + 2 \times 1}{\sqrt{8} \times \sqrt{11}} = 0.853$$

$$\theta \approx 31.4°$$

Which agrees well with the measured value and confirms the indexing assignment. Using the relation:

$$a = d\sqrt{H^2 + K^2 + L^2} = \frac{L\lambda}{R}\sqrt{H^2 + K^2 + L^2}$$

The lattice parameter is determined to be:

$$\mathrm{a} = 0.407\ nm$$